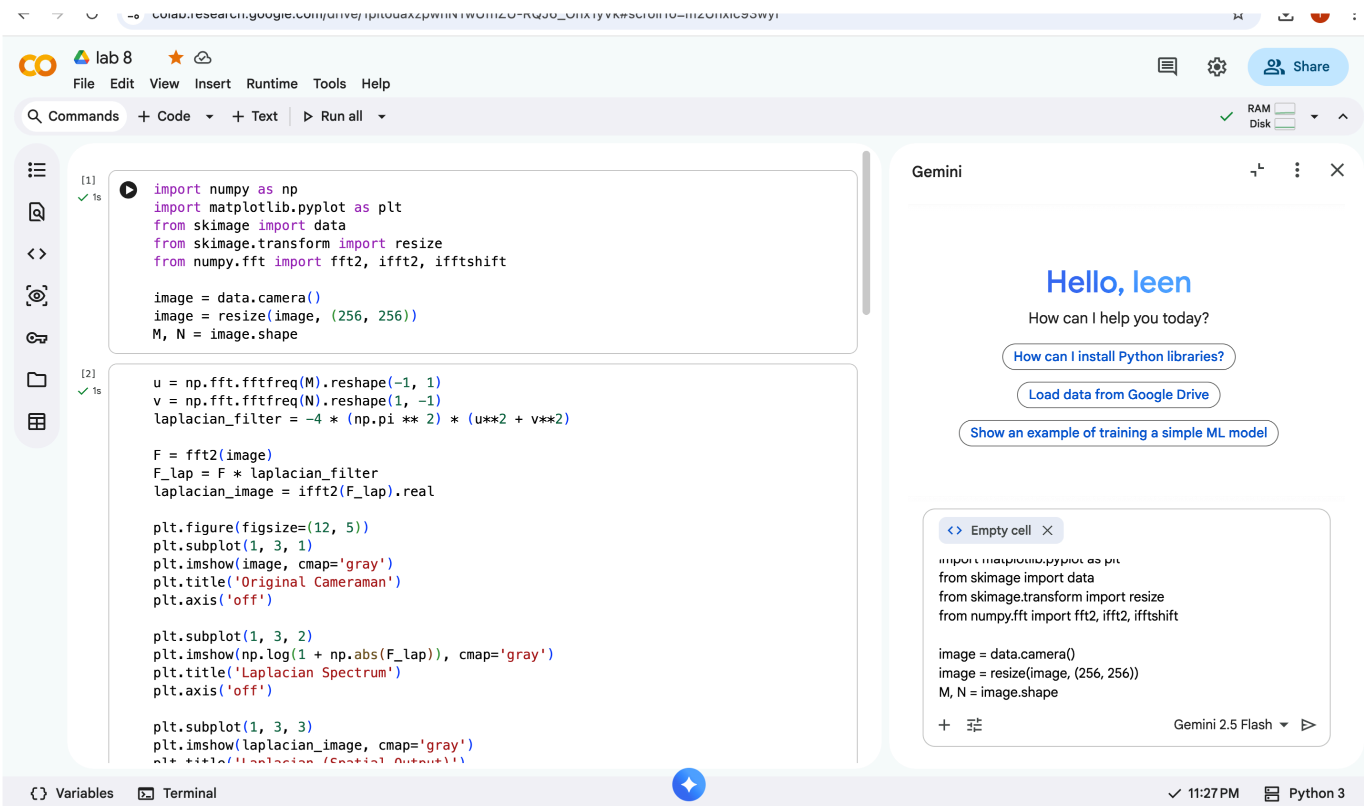
Image processing lab #8

Leen tariq batouk

2230002151



colab.research.google.com/drive/...

lab 8

File Edit View Insert Runtime Tools Help

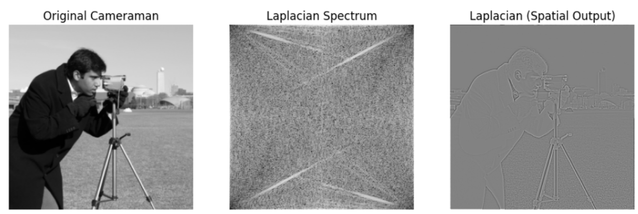
Commands + Code + Text Run all

[2]

```
plt.figure(figsize=(12, 5))
plt.subplot(1, 3, 1)
plt.imshow(image, cmap='gray')
plt.title('Original Cameraman')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(np.log(1 + np.abs(F_lap)), cmap='gray')
plt.title('Laplacian Spectrum')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(laplacian_image, cmap='gray')
plt.title('Laplacian (Spatial Output)')
plt.axis('off')
plt.show()
```



[3]

```
def center_embed_kernel(kernel, shape):
    padded = np.zeros(shape)
```

Gemini

Hello, leen

How can I help you today?

How can I install Python libraries?

Load data from Google Drive

Show an example of training a simple ML model

Empty cell

```
import matplotlib.pyplot as plt
from skimage import data
from skimage.transform import resize
from numpy.fft import fft2, ifft2, ifftshift

image = data.camera()
image = resize(image, (256, 256))
M, N = image.shape
```

Gemini 2.5 Flash

Variables Terminal 11:27 PM Python 3

colab.research.google.com/drive/...

lab 8

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

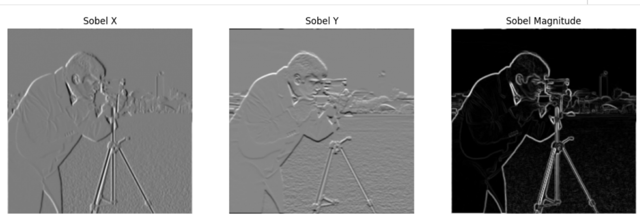
[3]

```
G_y = ifft2(F * H_y).real
sobel_magnitude = np.sqrt(G_x**2 + G_y**2)

plt.figure(figsize=(15, 5))
plt.subplot(1, 3, 1)
plt.imshow(G_x, cmap='gray')
plt.title('Sobel X')
plt.axis('off')

plt.subplot(1, 3, 2)
plt.imshow(G_y, cmap='gray')
plt.title('Sobel Y')
plt.axis('off')

plt.subplot(1, 3, 3)
plt.imshow(sobel_magnitude, cmap='gray')
plt.title('Sobel Magnitude')
plt.axis('off')
plt.show()
```



Gemini

Hello, leen

How can I help you today?

How can I install Python libraries?

Load data from Google Drive

Show an example of training a simple ML model

Empty cell

import matplotlib.pyplot as plt
from skimage import data
from skimage.transform import resize
from numpy.fft import fft2, ifft2, ifftshift

image = data.camera()
image = resize(image, (256, 256))
M, N = image.shape

Gemini 2.5 Flash

Variables Terminal 11:27 PM Python 3